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Study course: **Mathematics for Data and Resources Science (MDRS)**
Semester: **SS 2026**

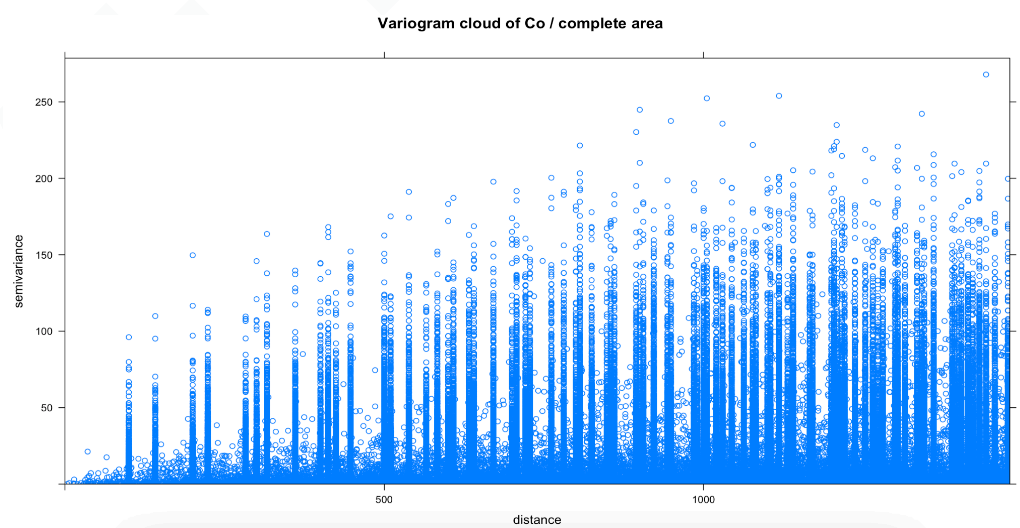
Geomodelling - Geostatistics for Natural Resource Evaluation
Report Variography

I) Task description

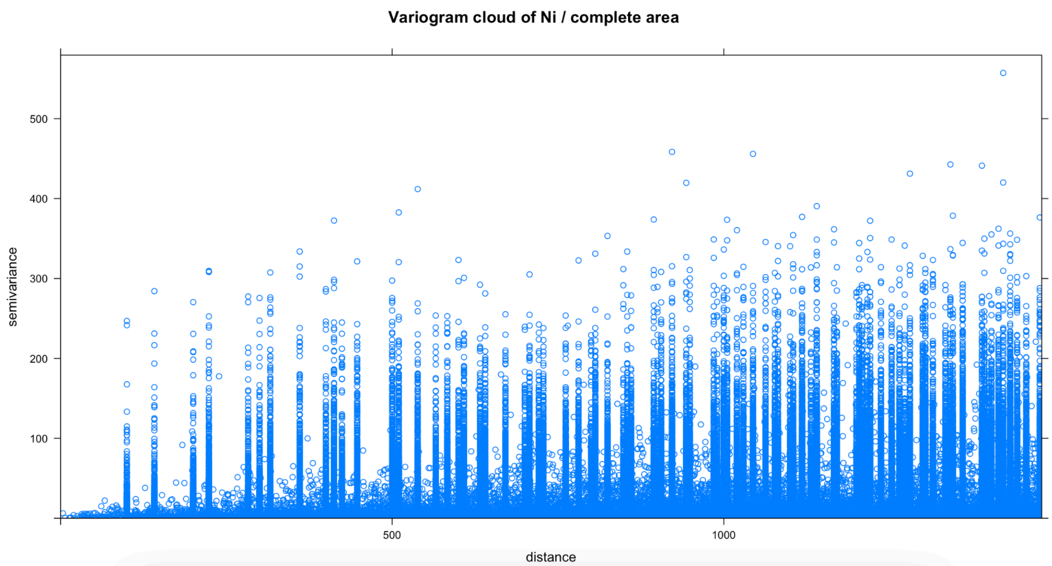
The aim of this report is to do a variogram analysis for the dataset **testData123.csv**. The analysis is done for the variables **Co** and **Ni**. First, I calculate the empirical variogram for the complete study area. Then the area will be divided into two parts and the variograms of both parts will be compared with the variogram of the complete area.
In this report I also check the variogram cloud, empirical variograms, possible nugget, sill and range values, fitted variogram models and possible anisotropy. The fitted variogram parameters are compared in a table. all calculations and plots are done with R.

II) Empirical variogram

The variogram clouds are very scattered. Most point pairs have low semivariance values, especially at short distances, but there are also several very high points above the main cloud. These high values are possible outliers pairs. The number of pairs is high over the whole plotted distance range, approximately up to 1400–1500.

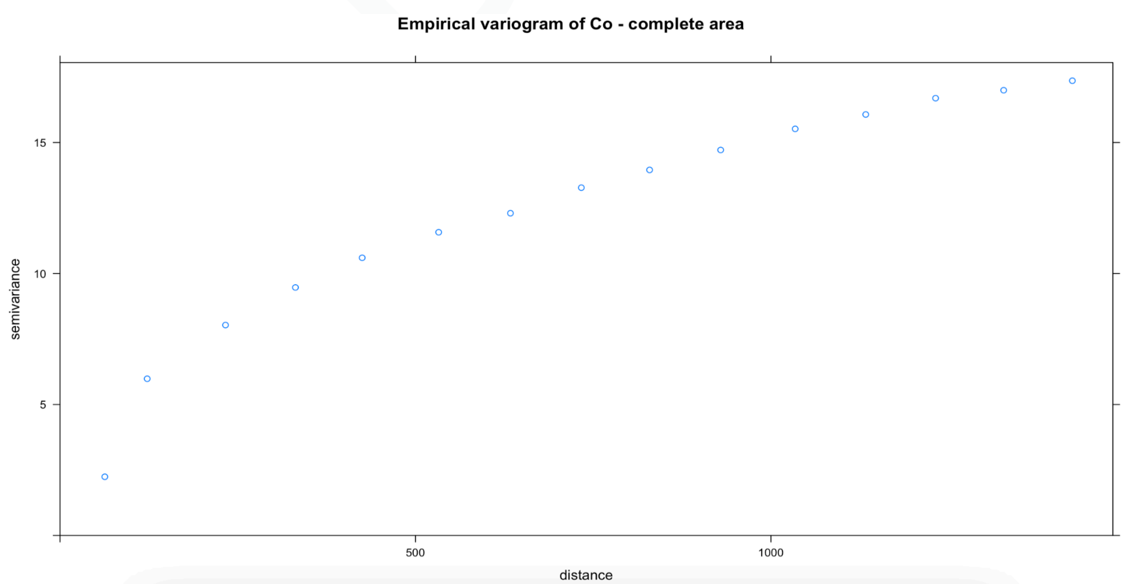


Variogram cloud of Co for the complete area

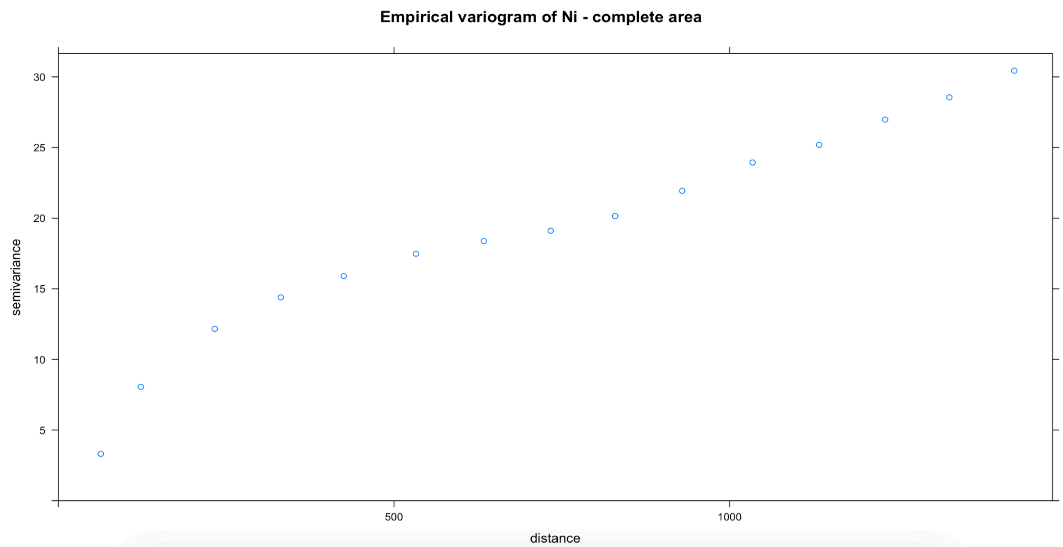


variogram cloud of Ni for the complete area

The empirical variograms of Co and Ni for the complete area show an increasing semivariance with increasing distance. So points close to each other are more similar than points farther away. For both variables, the variogram does not reach a very **clear flat sill** within the plotted distance. Therefore the **range** and **sill** can only be estimated from the empirical variogram.



Empirical variogram of Co



Empirical variogram of Ni

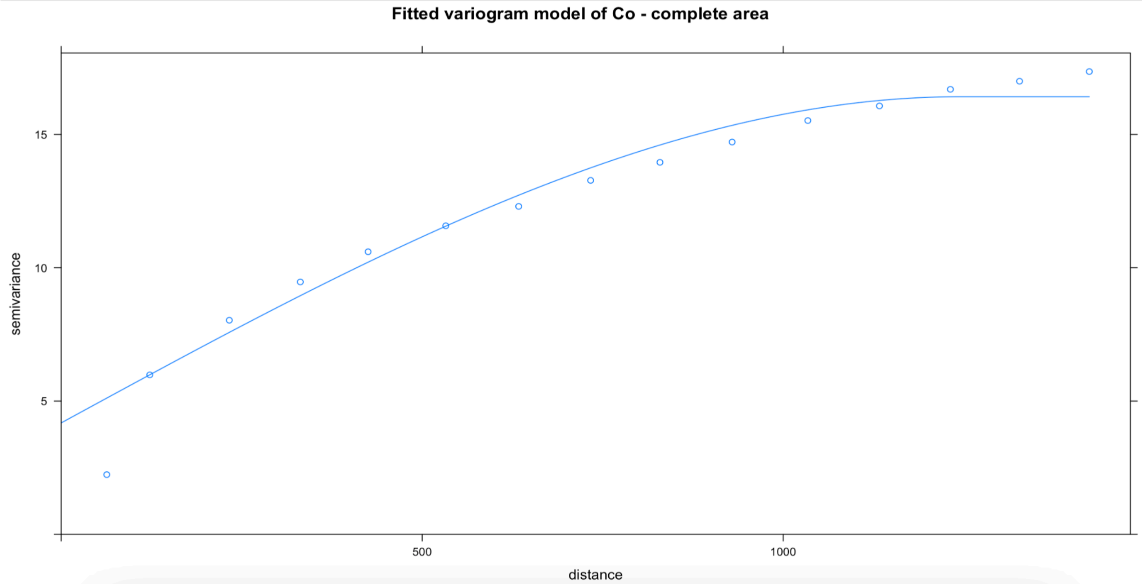
Both empirical variograms show increasing semivariance with increasing distance. Sample points close to each other are more similar than sample points far away from each other.

Co: the semivariance increases from **2.24** at a distance of **63**m to about **17.36** at distance of **1424**m. The empirical variogram comes near to the variance of Co which is about **18.27**. **it does not show a very clear sill**. So the nugget can be estimated between **1 and 2**, the sill at about **18**, and the range at about **1400**m or more.

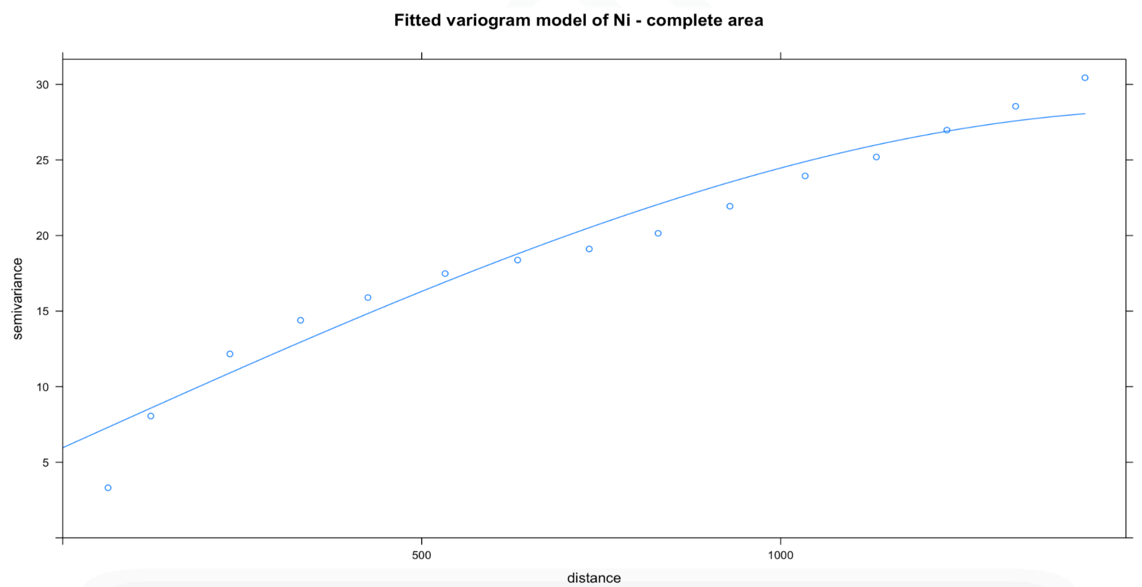
Ni: the semivariance increases from about **3.30** at a distance of **63**m to about **30.45** at a distance of **1424**m. The variance of Ni is **33.83** so the variogram is also close to the possible sill, but it is still increasing. Therefore, the nugget can be estimated between **2 and 3**, the sill between **32** and **34**, and the range at more than **1400**m.

The number of point pairs is sufficient in all distance groups. The first group already has 105 pairs and all other groups have more than thousand pairs. S**o, the empirical variograms can be used for the next modelling step.**

III) Variogram Modelling



Fitted spherical variogram model of **Co** for the complete area



Fitted spherical Variogram model of **Ni** for the complete area

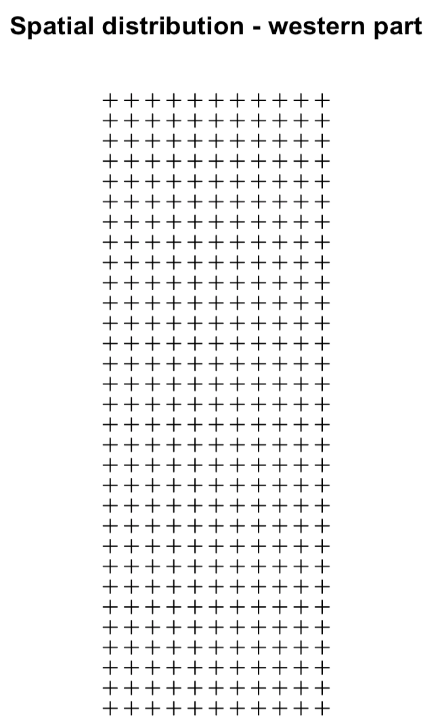
Variable	Model	Nugget	Partial sill	Total sill	Range
Co	Spherical	4.18	12.23	16.41	1243.66 m
Ni	Spherical	5.96	22.37	28.33	1566.91 m

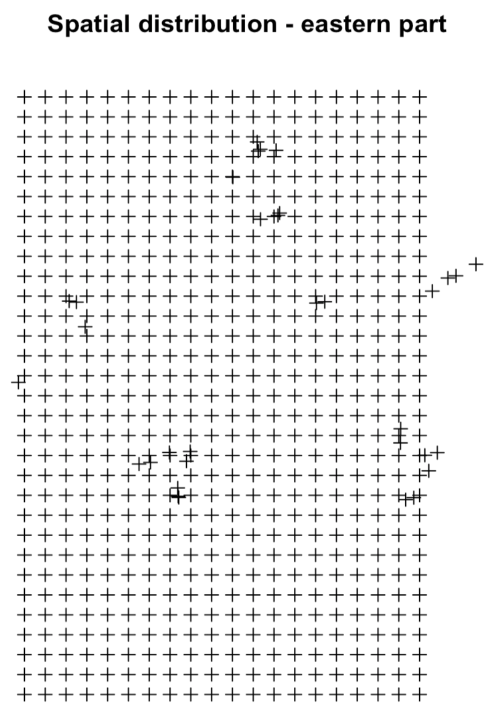
fitted variogram parameters for the complete area

A simple spherical variogram model was fitted to the empirical variograms of Co and Ni for the complete area. At this step an Empirical variogram shows an increase with distance and no need for another model can be seen from the first visual check.

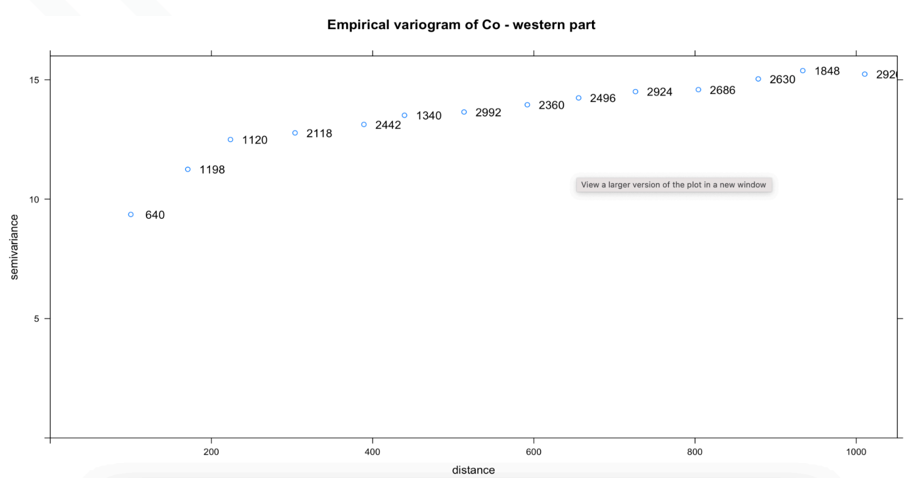
Co: the fitted spherical model has a nugget of **4.18**, a partial sill of **12.23** and a range of **1243.66** m. The total sill is therefore about **16.41**. The model follows an increasing shape of the empirical variogram, but the first empirical point is lower than the fitted curve. So the fit is not perfect.

Ni: the fitted spherical model has a nugget of **5.96**, a partial sill of **22.37** and a range of **1566.91**m. The total-sill is about **28.33**. The model also follows a trend of the Empirical variogram. Similar to Co, the first empirical point is lower than the fitted model, but the general shape is represented quite well. For the complete area, a simple spherical model seems usable for both variables.

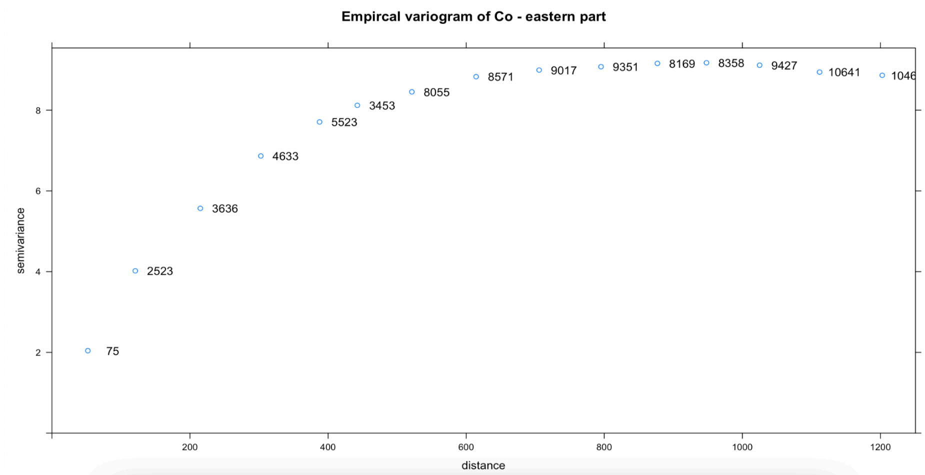




The study area was divided into a western part and an eastern part using the x-coordinate. The western part contains **341** points and the eastern part contains **653** points. Both parts have enough point pairs for the empirical variogram calculation. The first distance group already contains **640** pairs in the western part and **75** pairs in the eastern part.

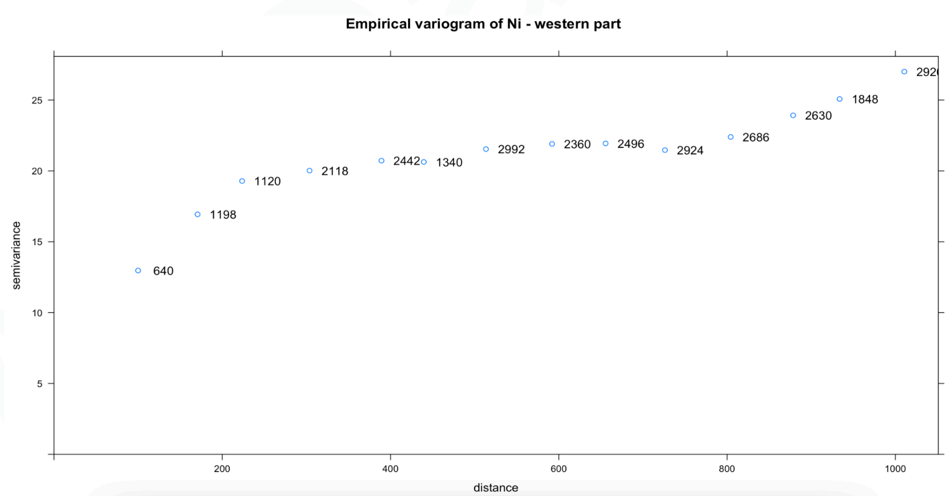


Empirical variogram of Co for the western part

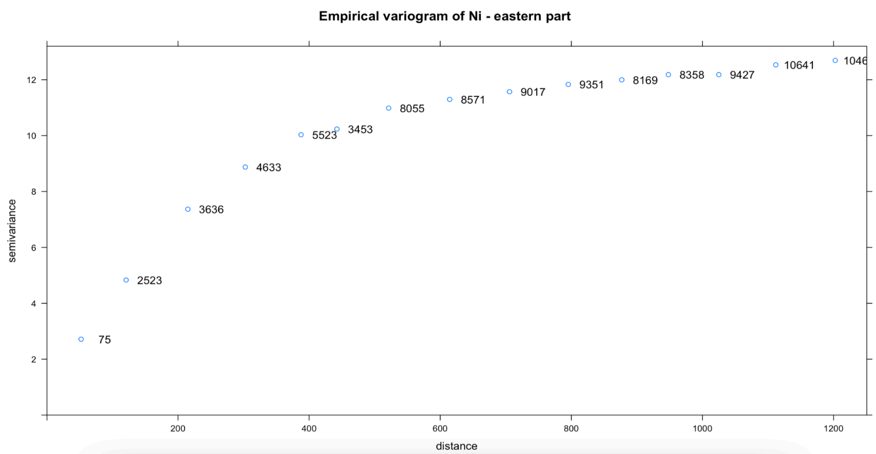


Empirical variogram of Co for the eastern part

Co: the western part has much more semivariance values than the eastern part. The western Co variogram starts at about **9.36** and increases to about **15**, while the eastern Co variogram starts at about **2.04** and reaches only about **9**. This means that **Co has stronger variability in the western** part.



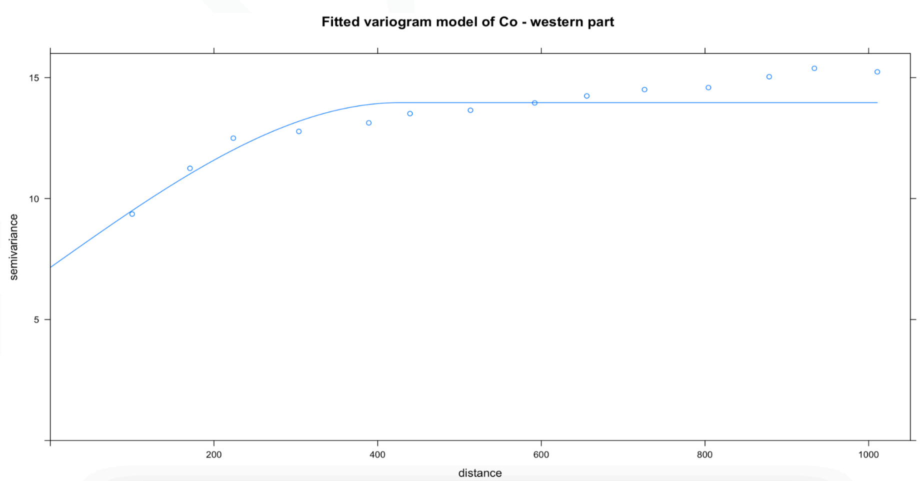
empirical variogram of Ni for the western part



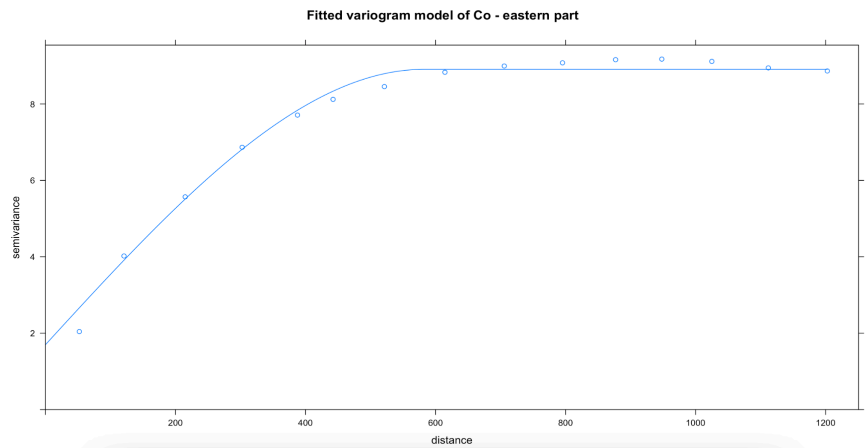
empirical variogram of Ni for the eastern part

Ni: the difference is also clear. The western Ni variogram starts at about **12.97** and increases to about **27**, while the eastern Ni variogram starts at about **2.72** and reaches only about **12 to 13**. Therefore, **Ni also shows stronger variability in the western** part.

Because the variograms of the western and eastern parts are clearly different, it would probably not be good to use only one variogram model for the whole area without checking the separate parts. For both parts, simple spherical variogram models were fitted for Co and Ni. The fitted parameters show that the two parts do not have the same variogram structure.

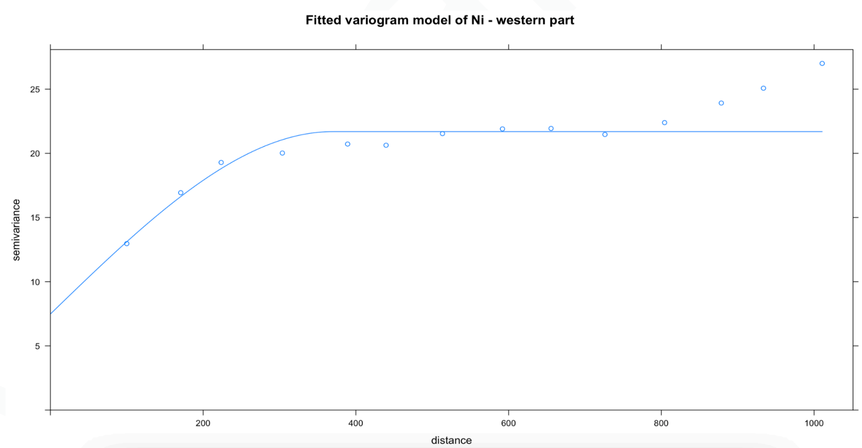


Fitted spherical variogram model of Co for the western part

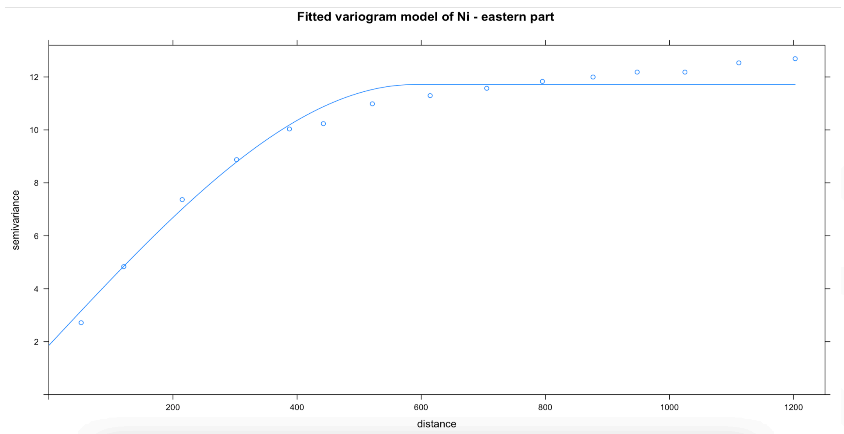


Fitted spherical variogram model of Co for the eastern part

Co: the western part has a nugget of **7.15**, a partial sill of **6.81** and a range of **427.21** m. The eastern part has a nugget of **1.70**, a partial sill of **7.21** and a range of **581.87**m. The complete area has a much larger range of **1243.66**m. This shows that the complete area model is different from the models of the two separate parts.



Fitted spherical variogram model of Ni for the western part



Fitted spherical variogram model of Ni for the eastern part

Ni:, the western part has a nugget of **7.48**, a partial sill of **14.21** and a range of **369.38** m. The eastern part has a nugget of **1.86**, a partial sill of **9.85** and a range of **588.32**m. Again, the complete area has a much larger range of **1566.91**m.

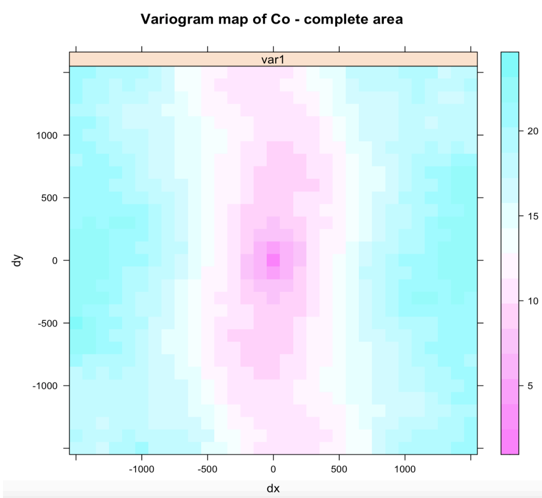
Variable	Area	Model	Nugget	Partial sill	Total sill	Range
Co	**Complete area**	Spherical	4.18	12.23	16.41	1243.66 m
Co	Western part	Spherical	7.15	6.81	13.97	427.21 m
Co	Eastern part	Spherical	1.70	7.21	8.91	581.87 m
Ni	**Complete area**	Spherical	5.96	22.37	28.33	1566.91 m
Ni	Western part	Spherical	7.48	14.21	21.70	369.38 m
Ni	Eastern part	Spherical	1.86	9.85	11.71	588.32 m

spherical variogram parameters for the **complete** area, **western** and **eastern** part

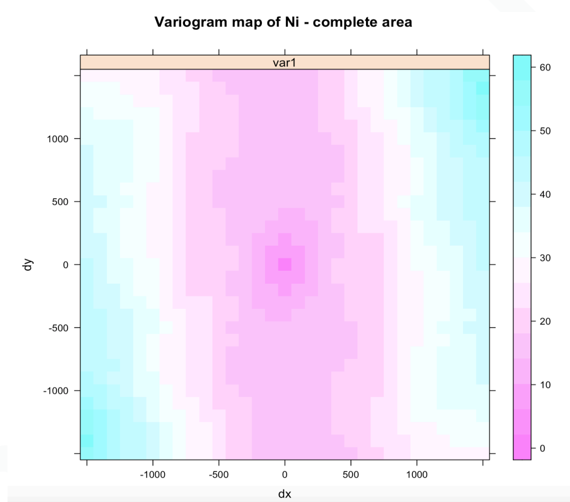
The western part has higher nugget values, which can show stronger short-distance variation or local variability. The eastern part has lower nugget and lower total sill values, so it seems spatially smoother. Because the parameters are clearly different, I would not use exactly the same variogram model for the complete area and for the two separate parts.

IV) Anisotropy

The anisotropy was checked using variogram maps and directional variograms for the complete area. The variogram maps of Co and Ni show that the semivariance is not the same in all directions. Therefore, a directional dependency is visible.



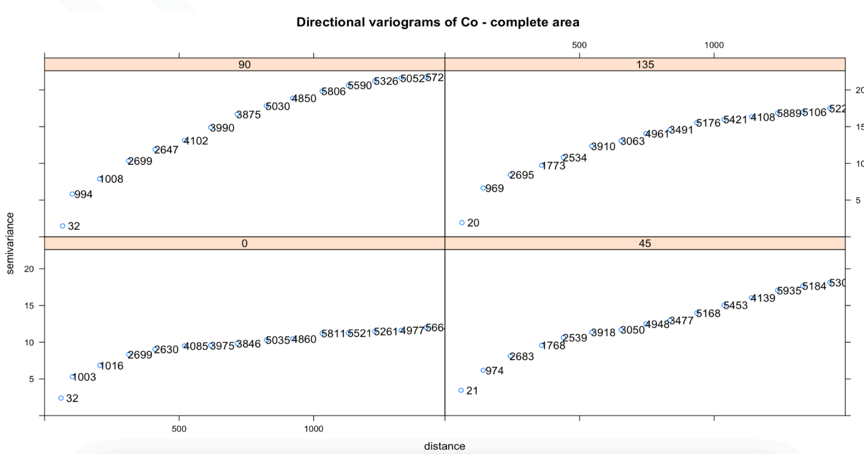
Variogram map of Co



Variogram map of Ni

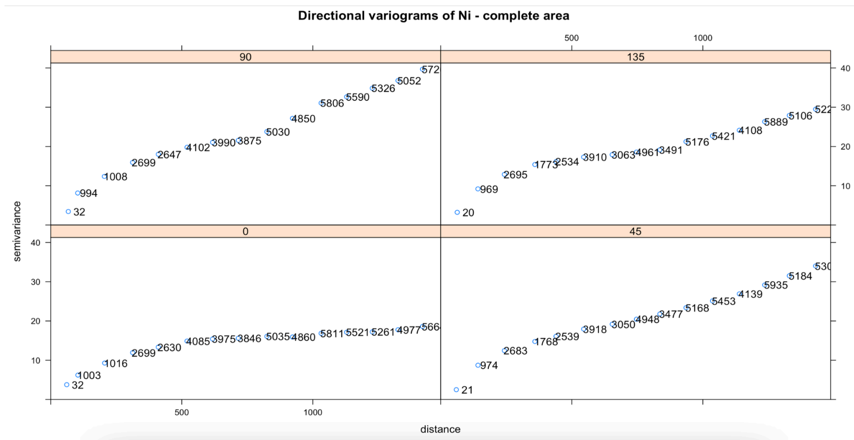
The directional variograms were calculated for the directions 0°, 45°, 90° and 135°.

Co: the 0° direction has lower semivariance values than the 90° direction. At the largest distance, the 0° direction reaches about 12, while the 90° direction reaches about 21.75. This means that Co has stronger spatial continuity in the 0° direction and weaker continuity in the 90° direction.



Directional variograms of Co

Ni: the directional difference is even stronger. The 0° direction reaches about 18.60 at the largest distance, while the 90° direction reaches about 39.69. This shows a clear anisotropic behaviour for Ni.



Directional variograms of Ni

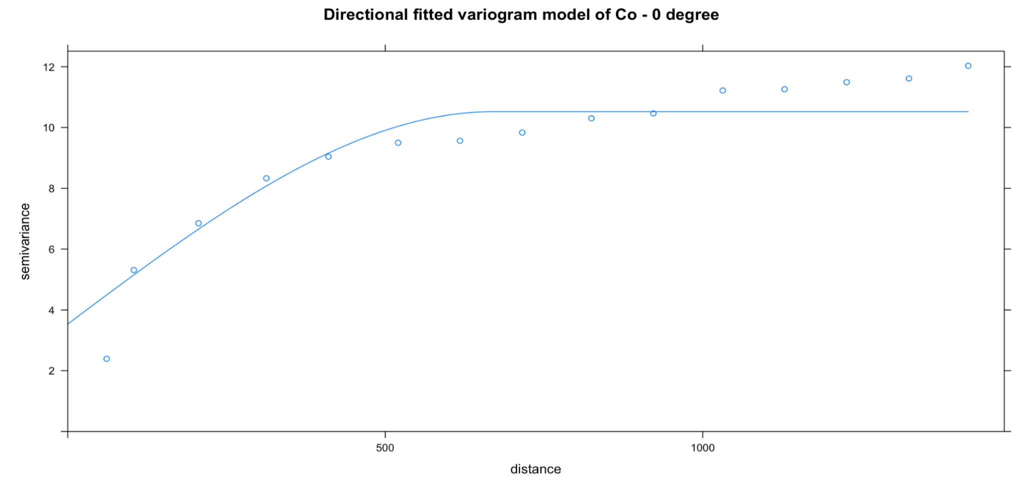
Most distance groups have enough point pairs. Some first distance groups, especially in the diagonal directions, have only about 20 to 21 pairs, so these first values should be interpreted carefully. However, the other distance groups contain many more point pairs, so the directional variograms are still useful for visual interpretation.

Overall, the data show anisotropy. The 0° and 90° directions are clearly different, so an isotropic variogram model may not describe the spatial structure completely.

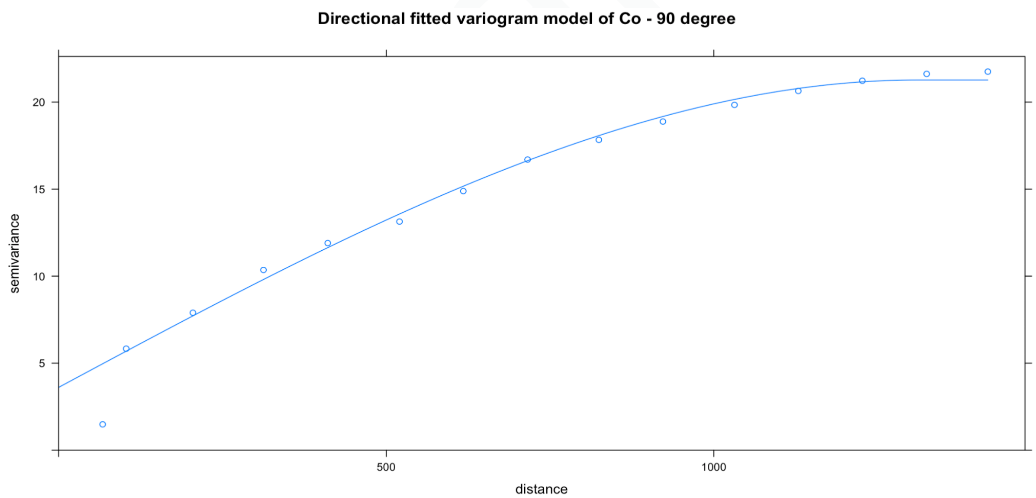
Variable	Direction	Model	Nugget	Partial sill	Total sill	Range
Co	0°	Spherical	3.54	6.99	10.52	669.29 m
Co	90°	Spherical	3.61	17.66	21.27	1311.00 m
Ni	0°	Spherical	3.21	13.17	16.37	685.76 m
Ni	90°	Spherical	5.83	37.37	43.20	2124.28 m

Directional variogram model parameters for 0° and 90° directions

Because the directional variograms showed clear differences between 0° and 90°, directional spherical variogram models were fitted for these two directions.

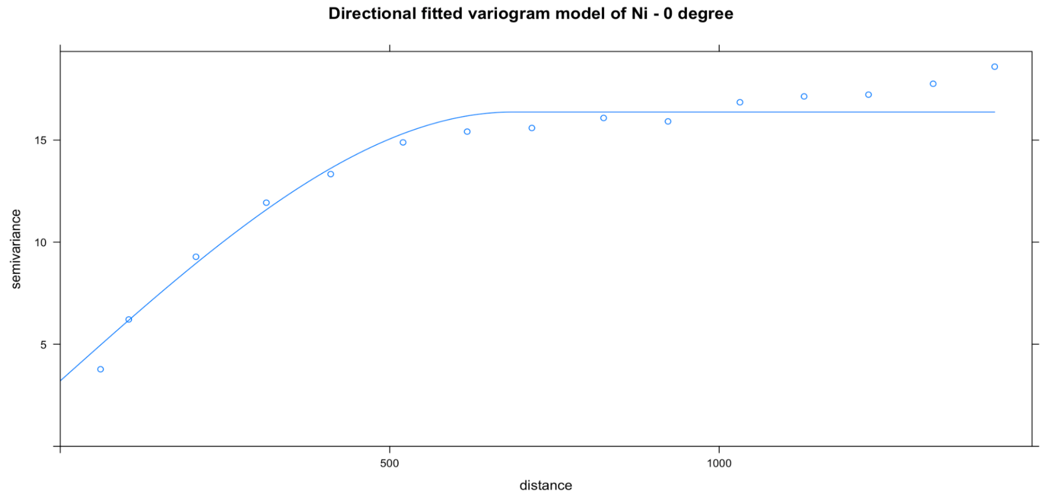


Directional fitted variogram model of Co in 0° direction

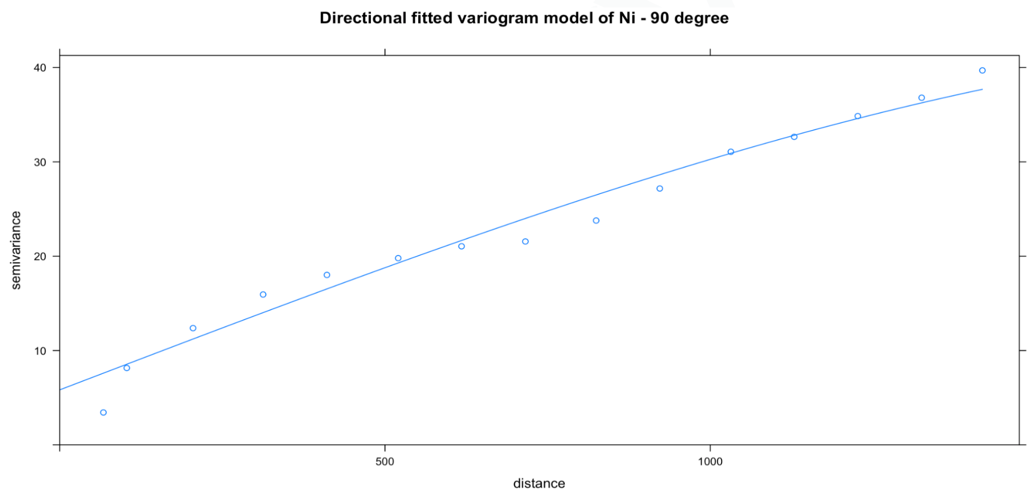


Directional fitted variogram model of Co in 90° direction

Co: the fitted model in the 0° direction has a nugget of 3.54, a partial sill of 6.99 and a range of 669.29 m. In the 90° direction, the nugget is 3.61, the partial sill is 17.66 and the range is 1311.00 m. This shows that Co has different spatial behaviour in the two directions.



Directional fitted variogram model of Ni in 0° direction



Directional fitted variogram model of Ni in 90° direction

Ni: the fitted model in the 0° direction has a nugget of 3.21, a partial sill of 13.17 and a range of 685.76 m. In the 90° direction, the nugget is 5.83, the partial sill is 37.37 and the range is 2124.28 m. The difference between the two directions is even stronger for Ni than for Co.

These results confirm that the data are anisotropic. The 90° direction has a much larger range and total sill than the 0° direction. Therefore, an isotropic variogram model is useful as a first model, but it does not describe all directional behaviour of the data.

V) Results

In this variogram analysis, empirical variograms and fitted spherical variogram models were calculated for Co and Ni. The analysis was done for the complete area and also separately for the western and eastern parts of the study area.

For both Co and Ni, the empirical variograms of the complete area show **increasing semivariance with increasing distance**. This means that nearby points are more similar than points far away from each other. **A simple spherical model was fitted for both variables**. The spherical model describes the general structure of the empirical variograms, so it can be used as a first model.

However, the comparison between the western and eastern parts shows clear differences. The western part has higher nugget values and higher semivariance values than the eastern part. The eastern part has lower total sill values and seems spatially smoother. **Therefore, the two parts do not have exactly the same variogram structure**.

The anisotropy analysis also shows directional dependency. The variogram maps and directional variograms show that the 0° and 90° directions are different. For both Co and Ni, the 90° direction has a larger range and larger total sill than the 0° direction. This means that **an isotropic model does not describe all spatial behaviour completely**.

For further analysis, I would use spherical variogram models, but I would not use one single isotropic model for all cases. If the complete area has to be analysed as one dataset, the fitted spherical model of the complete area can be used as a simple first model. But because the western and eastern parts show different variogram parameters and because anisotropy is visible, separate variogram models for the two parts or directional variogram models would be more appropriate for a more detailed geostatistical analysis.

Overall, Co and Ni both show spatial dependence, different behaviour between the western and eastern parts, and anisotropy. Therefore, the spherical model is acceptable, but the final choice should consider the area division and directional dependency.